



UNIVERSITY OF PISA

DEPARTMENT OF COMPUTER SCIENCE

Master's Degree in Computer Science

AMM Scheduler:

**Design and Implementation of a Runtime
Scheduling System for Asymmetric Heterogeneous
Computing**

Supervisor:

Prof. Marco Danelutto

Candidate:

Simone Stanganini

ACADEMIC YEAR 2024/2025

Contents

1	Introduction	5
1.1	Contest	5
1.2	Objectives	6
1.3	Thesis Structure	6
2	Tools and Technologies	7
2.1	Experimental Platform	7
2.1.1	The SoC: Intel Core Ultra 9 185H	8
2.1.2	NVIDIA GeForce RTX 4060	9
2.2	CUDA Toolkit	9
2.3	SYCL	11
2.4	Openvino	12
2.5	OpenBlass	13
2.6	C/C++	14
2.7	Cmake	15
3	Logical Design	17
3.1	Preliminary Phase	18
3.1.1	Studying the Common Patterns	18
3.1.2	Benchmarking	19
3.2	AMM Scheduler	20
4	Implementation	25
4.1	Project Structure and Compilation	26
4.1.1	Compilation Process	27

CONTENTS	4
4.2 Accelerators Abstraction Layers	33
4.2.1 Cuda	33
5 Results	39
A Questa è un'appendice	41

Chapter 1

Introduction

1.1 Contest

Until a few years ago, performance improvements depended mainly on increasing CPU clock speeds, today the approach has changed completely, shifting towards hardware specialization.

Modern computers, from servers to embedded SoCs, don't rely on a single processor anymore, instead they integrate different specific accelerators:

- **CPU**: Optimized for sequential tasks and low latency.
- **GPU**: Available as integrated (iGPU) or discrete (dGPU), in a lot of cases both at the same time, designed for massive parallel computing.
- **NPU** (Neural Processing Unit): A recently introduced unit, specialized in speeding up matrix operations for AI workloads.

This variety of hardware has created a gap with the software, while hardware offers different resources for different workloads, traditional software tends to be static. It usually assigns tasks to a single device, often just the CPU or GPU, and ignores the other available resources.

The result is an inefficient use of the system. We could get more throughput with the same efficiency, or be more efficient at the same speed, so the current challenge is *orchestration*: we need a system that can dynamically map each task to the right accelerator, using the entire heterogeneous topology of the system.

1.2 Objectives

The main goal of this thesis was to design and implement a scheduler capable of dynamically assigning computational tasks, specifically streams of matrix multiplications, to the different accelerators found in modern architectures.

The work started with an essential preliminary phase, I focused on integrating and studying different compute backends, such as CUDA, SYCL, OpenVINO. This step was useful to create a uniform abstraction layer over the hardware and to verify the technical feasibility of the system.

After this initial phase, the work moved on to the actual development of the scheduler, the core of this project focused on the asymmetric orchestration logic, designed to decide in real-time which resource to use for each specific workload for maximize utilization and efficiency.

Finally, the system was validated with experimental results on a complex hybrid architecture, this setup included an Intel multicore CPU (with integrated GPU and NPU) and a discrete NVIDIA GPU. The tests used to verify the operation were streams of identical matrices, streams of heterogeneous matrices, and splitting larger tasks across multiple accelerators.

1.3 Thesis Structure

The work is organized into the following chapters:

- **Chapter 2:** Introduces the hardware technologies (CPU, GPU, NPU), the software backends used (CUDA, SYCL, OpenVINO, OpenBLAS) and the software stack used for this project (C, C++, Cmake).
- **Chapter 3:** Describes the logical design, divided into two phases: the creation of the benchmarking environment and the subsequent design of the scheduler.
- **Chapter 4:** Details the code implementation, it covers the wrappers for hardware abstraction, the essential data structures for the scheduler (ringbuffer and performance map), and the techniques used for task splitting, as well as the project compilation procedures and details.
- **Chapter 5:** Analyzes the experimental results, validating the effectiveness of the scheduler on different workloads.
- **Chapter 6:** Draws conclusions on the work done and suggests possible future developments.

Chapter 2

Tools and Technologies

This chapter presents the hardware and software resources used to develop and validate the system. The choice to work on a hybrid and highly heterogeneous architecture was made to test the scheduler in a real-world scenario, and in this scenario, devices with different performance characteristics and programming models have to work together.

2.1 Experimental Platform

All development and benchmarking activities were done on a mobile workstation Lenovo Yoga Pro 9i gen 9. This platform represents a modern high-performance "Edge" computing node well, it integrates all the types of accelerators studied into a single physical system: multicore CPU, NPU, integrated GPU, and discrete GPU, and having these components available at the same time allowed to emulate, within a single node, the orchestration issues of heterogeneous distributed systems.

2.1.1 The SoC: Intel Core Ultra 9 185H

The core of the system is the Intel Core Ultra 9 185H processor, based on the "Meteor Lake" architecture [2]. This component marks a significant change from traditional monolithic CPUs. It uses a disaggregated design (tiled architecture) that combines different specialized processing units:



Figure 2.1: Intel Ultra Logo

- **Host CPU:** This cpu is made of 16 hybrid cores, 6 Performance-cores, 8 Efficient-cores, and 2 Low-Power-cores for efficiency. In this project this unit acts both as host and accelerator, it runs the scheduler code and does some computation through the OpenBlass library (2.5).
- **NPU:** The Ultra 9 includes a native Neural Processing Unit. This is a low-power accelerator designed specifically for neural network inference. Its compute acceleration is facilitated by Neural Compute Engines, which consist of hardware acceleration blocks for AI operations like Matrix Multiplication and Convolution, alongside Streaming Hybrid Architecture Vector Engines for general computing tasks [4]. In this project the NPU is used via the OpenVINO 2.4 backend to run Matrix Multiplication.
- **iGPU:** Intel Arc Graphics, integrated into the SoC, this graphics unit based on Xe-LPG architecture [1], offers parallel computing capabilities with direct access to system memory, in this project this accellerator has been used with the SYCL library (2.3) as the backend to leverage the full performance of this accellerator.

2.1.2 NVIDIA GeForce RTX 4060



Figure 2.2: Nvidia Rtx Logo

To complement the Intel SoC the system is equipped with a discrete NVIDIA GeForce RTX 4060 Laptop GPU. This device is based on the Ada Lovelace architecture [8] and acts as the high-throughput accelerator of the system.

Unlike the integrated GPU this card features 3072 CUDA Cores for general parallel processing and 96 Fourth-Generation Tensor Cores. The Tensor Cores are specifically designed to accelerate dense matrix multiplications with 16 or even 8 bit operations.

The device utilizes 8 GB of GDDR6 VRAM with a 128-bit memory interface, this dedicated memory provides high bandwidth but requires explicit data transfer over the PCIe bus.

Within the scheduler this device is managed via the CUDA backend 2.2.

2.2 CUDA Toolkit



Figure 2.3: Nvidia Cuda Logo

To exploit the computational capabilities of the discrete NVIDIA GPU the system utilizes CUDA (Compute Unified Device Architecture). This parallel computing platform and programming model developed by NVIDIA that enables dramatic increases in computing performance by harnessing the power of the GPU. It allows developers to accelerate compute-intensive applications using C, C++, and Fortran, and is widely adopted in fields such as deep learning, scientific computing, and high-performance computing (HPC) [10].

In this project CUDA is the tool chosen to extract maximum throughput from the hardware 2.1.2. The implementation relies on specific libraries and mechanisms to optimize the execution of matrix multiplication streams.

- **cuBLAS and Tensor Cores:** to perform linear algebra operations we adopt cuBLAS (CUDA Basic Linear Algebra Subroutines). This library is the GPU-accelerated version of the standard BLAS specification and provides highly optimized implementations of matrix operations, specifically, cuBLAS allows the utilization of Tensor Cores, specialized hardware units designed to accelerate mixed-precision matrix multiplication (Float16 and Int8). This significantly increases the theoretical throughput compared to standard floating-point units [9].
- **CUDA Streams:** To handle the latency introduced by data transfers between the CPU and the GPU the system utilizes CUDA Streams. A stream is a sequence of operations that execute in issue-order on the GPU, by employing multiple streams it is possible to achieve concurrency: while one stream is transferring data over the PCIe bus another stream can execute a computational kernel [13]. This mechanism, known as "latency hiding", is fundamental for real-time processing and ensures that the GPU compute units remain active as much as possible.
- **NVIDIA Management Library (NVML):** For the analysis of energy efficiency the project integrates the NVIDIA Management Library (NVML). This serves as a monitoring tool that provides direct access to the GPU hardware sensors, it allows the retrieval of real-time metrics such as power usage (in milliJoule) and temperature without requiring external physical probes [11].

2.3 SYCL

While CUDA provides maximum performance for NVIDIA hardware, it requires a proprietary programming model, and to address the need for portability and to fully exploit the Intel hardware present in the system, the project adopts the SYCL standard through the Intel oneAPI toolkit.

- **The SYCL Standard:**



Figure 2.4: SYCL Logo

SYCL is an open, royalty-free standard managed by the Khronos Group. It enables high-performance computing on heterogeneous accelerators (such as CPUs, GPUs, and FPGAs) using standard ISO C++ [15]. As defined in the official specification, SYCL provides an abstraction layer that allows developers to write code for heterogeneous processors using a "single-source" style. This means that host code on the CPU and the device code on the accelerator are written in the same file, utilizing standard C++ templates and lambda functions. This approach eliminates the complexity of managing separate source files for different architectures.

- **Intel oneAPI:**

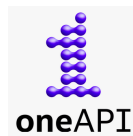


Figure 2.5: OneApi Logo

SYCL requires a specific software development kit (SDK) to be compiled and executed. For this thesis, the Intel oneAPI Base Toolkit was selected [3]. oneAPI is Intel's unified programming model designed to simplify development across diverse architectures, it implements the SYCL standard via the `icpx` compiler, and provides a comprehensive set of libraries optimized for a large set of devices. The choice of this toolkit allows the project to target the available Intel hardware efficiently. Moreover this choice guarantees that the codebase could seamlessly target accellerator from othe vendor like AMD without having to rewrite the code.

2.4 Openvino



Figure 2.6: OpenVino Logo

Developed by Intel, this open-source toolkit is designed to optimize and deploy deep learning models across various types of hardware, from edge devices to cloud servers [5].

The main function of the library is that it acts as a translator, it takes an AI model and converts it into a specific format that the target hardware understands. This is very useful because it allow the program to run on different components like the CPU, NPU or the integrated graphics without needing to rewrite the code for each specific accellerator.

The toolkit automatically handles the low-level details to ensure the task runs as fast as possible in the chosen device. This efficiency is achieved through the model optimizer, as described in the documentation, this tool analyzes the computational operations and simplifies them (for example, by merging multiple mathematical steps into one) before execution.

The result is a hardware-agnostic format called intermediate representation, IR, which allows the runtme to map the operations efficiently to the available resources [5].

In this project OpenVINO was essential for targeting the specific NPU found in the system 2.1.1. Since the NPU cannot run standard C++ code directly, OpenVINO acts as the bridge between the scheduler and the hardware, specifically, it was implemented to execute Matrix Multiplication tasks on the NPU. This allowed the system to offload these calculations to the neural accelerator, taking advantage of its high energy efficiency.

2.5 OpenBlass



Figure 2.7: OpenBlass Logo

To enable efficient processing on the Host CPU, the project integrates OpenBLAS. OpenBLAS is an optimized, open-source implementation of the BLAS (Basic Linear Algebra Subprograms) API.

Historically, the project emerged as a successor to GotoBLAS2, a highly regarded library developed by Kazushige Goto, then after the original development ceased, the OpenBLAS community took over to maintain the code and expand support for modern architectures [16].

The library achieves high performance through a mechanism called runtime processor detection. OpenBLAS first identifies the CPU architecture and automatically selects the most optimized functions for that specific hardware, these functions are often implemented in assembly language to leverage SIMD instructions, such as AVX2 or AVX-512, which allow the processor to handle multiple data points simultaneously. Additionally, the library automatically handles multi-threading to distribute the computational load across all available processor cores[12].

In this project this library allow the Host CPU to be treated as an additional accelerator, so rather than limiting the CPU solely to orchestration tasks, OpenBLAS enables the scheduler to use some of the processor cores as a worker node. More details on this will be provided in the Implementation Chapter 4.

2.6 C/C++



Figure 2.8: C and C++ Logo

The C and C++ programming languages represent two of the most influential and widely used tools in the history of computer science. Known for their performance, versatility, and low-level control, they are employed in a vast range of sectors, from operating systems to high-performance applications.

Below is a brief overview of both languages:

The C Language:

- Developed in the 1970s by Dennis Ritchie, it was one of the first high-level languages in history.
- it is renowned for its simplicity, efficiency, and direct access to memory, making it ideal for writing highly optimized code [6].

The C++ Language:

- C++ is an extension of C developed in the 1980s by Bjarne Stroustrup. Among its key features is the support for Object-Oriented Programming (OOP), which allows for the creation of more structured and modular software compared to procedural C [14].
- It provides a rich Standard Library (STL) and is the industry standard for performance-critical applications.

In the context of this project, C plays a critical architectural role. It was utilized to define the Application Binary Interface (ABI) between the scheduler and the various hardware wrappers. Instead C++ serves as the primary development language. Its adoption was mandatory as it is the native language required by the SDKs of all the utilized frameworks. It is used to implement the complex logic of the scheduler and the data structures necessary for task management. Further details on this will be provided in the Implementation Chapter 4.

2.7 Cmake



Figure 2.9: Cmake Logo

To manage the compilation process of such a complex architecture, the project utilizes CMake (Cross-Platform Make). CMake is an open-source family of tools designed to build, test, and package software. It was originally created by Kitware in 2000 to address the need for a powerful build environment capable of supporting cross-platform development [7].

CMake does not build the software directly, instead it acts as a generator: it reads configuration files named CMakeLists.txt, and generates standard build files for the native toolchain of the operating system, such as Makefiles for Linux or Visual Studio projects for Windows. This abstraction allows developers to describe how the project should be built in a way that is independent of the specific compiler being used.

In this thesis its primary function is to manage the complexity of dynamic linking: since the project depends on large external frameworks (CUDA, OpenVINO, OneAPI, OpenBlass), CMake is configured to locate and link these as shared libraries. This ensures that the executable remains lightweight and that dependencies are resolved correctly at runtime.

Additionally, it enables a modular structure where specific hardware backends can be included or excluded via simple configuration flags.

Chapter 3

Logical Design

This chapter describes the logical structure of the project, the design process was divided into two main phases:

- The first part of the chapter (3.1) presents the Independent Benchmarks. Before developing the centralized scheduler, standalone applications were created for each accelerator. This phase was essential to verify the feasibility of the project and to understand how to effectively control the specific hardware using the selected libraries.
- The second part (3.2) provides a comprehensive overview of the AMM Scheduler. This section describes the high-level operation of all the components of the scheduler, it illustrates the architectural design of the main project and explains how the various logical blocks interact to enable the scheduler's functionality, detailing the specific operational logic that governs the system.

3.1 Preliminary Phase

Before designing the centralized scheduler, the project required a preliminary phase focused on the independent analysis of each software stack. Three standalone applications were developed, targeting the NVIDIA GPU, the Intel iGPU, and the NPU respectively. The primary objective of this phase to gain a deep understanding of the specific libraries (CUDA, SYCL, and OpenVINO) and their interaction with the hardware.

It is important to note that the OpenBLAS backend for the CPU was not included in this preliminary phase. This component was introduced directly during the development of the scheduler (described in the next section 3.2) with a specific goal: to maximize the degree of heterogeneity of the system, by treating the CPU as an additional accelerator rather than just a coordinator. Moreover unlike the GPU and NPU, which required a specific study of their asynchronous drivers and memory models, the CPU integration relies on standard synchronous function calls, consequently, a dedicated standalone analysis deemed unnecessary. The technical details of this integration will be discussed in the implementation chapter (4).

3.1.1 Studying the Common Patterns

A crucial goal of this stage was to gain a deep, hands-on understanding of the specific libraries. Since technologies like CUDA, SYCL, and OpenVINO manage hardware in fundamentally different ways, so a first approach phase was necessary to see how their specific mechanisms works and the drivers interactions.

This distinction is particularly evident in the case of OpenVINO, while CUDA and SYCL follow a traditional *kernel launch* paradigm, the NPU is designed exclusively for deep learning graphs, not for general-purpose linear algebra, consequently, to perform a standard matrix multiplication (GEMM) on this accelerator, a workaround was required, instead of invoking a mathematical function, the operation had to be encapsulated within a synthetic neural network layer, effectively "tricking" the driver into treating a raw calculation that we wanted to accomplish as an inference task (more on this in section ??).

Despite these architectural differences, this preliminary study revealed a recurring pattern, the execution flow for every accelerator whether running a kernel or a simulated inference relies on four identical logical steps:

- **Context Initialization:** setting up the driver handles like context, queue, and so on.
- **Memory Management:** allocating buffers and managing host-to-device transfers.
- **Kernel Execution:** triggering the asynchronous computation.
- **Synchronization:** waiting for the hardware to complete the task.

Recognizing this common structure for each library was fundamental for the design of the Abstraction Layer used in the scheduler.

3.1.2 Benchmarking

During this phase, benchmarking routines were implemented within each of the standalone applications. However, their purpose was distinct from the final experimental results presented in later chapters, in this context measuring execution time and power consumption served as a development tool to:

- Verify the correctness of the implementation (e.g., ensuring the code was actually running on the accelerator and not falling back to the CPU).
- Analyze the runtime behavior of the drivers, and identify architectural bottlenecks (such as PCIe transfer overheads or compilation latencies).

However, it must be noted that the performance profile observed in these isolated environments proved to be different from the behavior within the full scheduler. The concurrent execution of different workloads and the increased stress on system resources exposed new bottlenecks that were not visible during the standalone tests, consequently, the backend implementations were significantly evolved during the integration phase, and specific structures were introduced to mitigate these latencies. These optimizations and the specific issues encountered under high-load scenarios will be analyzed in detail in the Implementation Chapter (4).

3.2 AMM Scheduler

The core of the project is the AMM Scheduler, that stands for Asymetric Matrix Multiplication Scheduler.

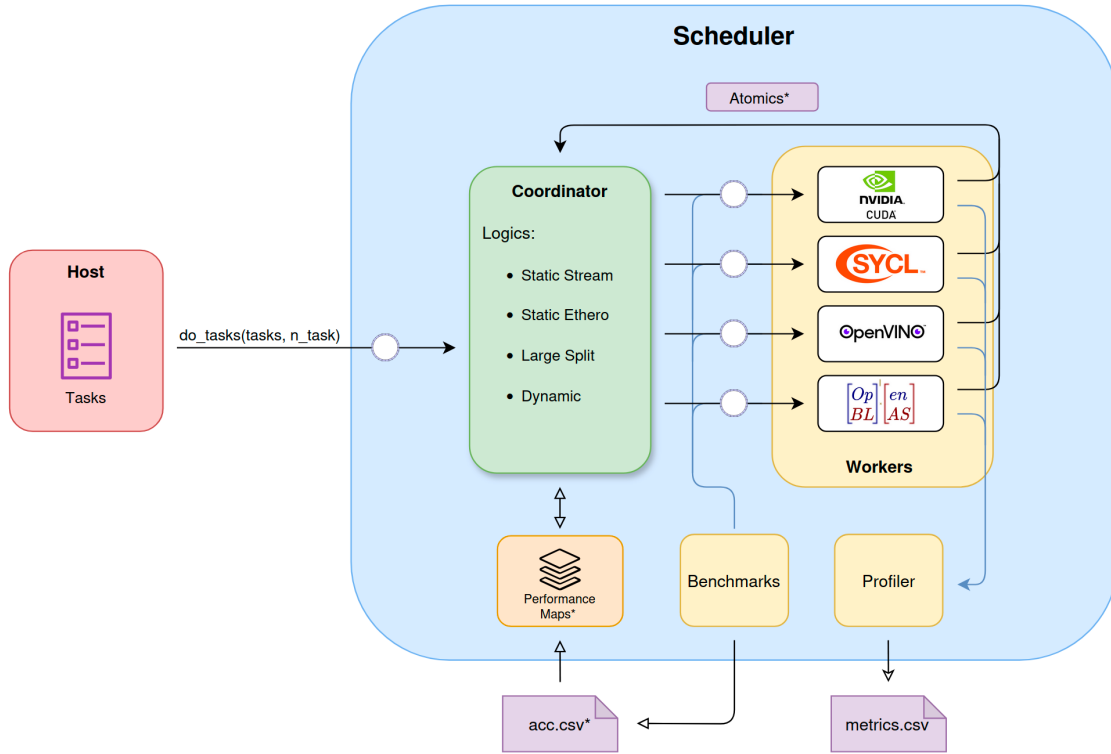


Figure 3.1: AMM Scheduler Components

As illustrated in figure 3.1, the architecture implements the producer and consumer paradigm.

In this model, the host application acts as the producer, generating computational tasks for the scheduler, that functions as the consumer, processing these requests asynchronously. The execution logic follows the farm pattern: a central coordinator retrieves the tasks and distributes them among the available worker nodes. To enable parallel execution, the coordinator and each worker operate as independent threads.

As shown in the diagram, Workers communicate with the Coordinator using atomic variables. This mechanism signals when a Worker is idle or updates the count of completed tasks, this ensures that the scheduler's wait function can correctly determine when all tasks are finished.

Now we will examine each component in the diagram and explain a bit deeper its function:

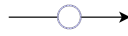


Figure 3.2: Ring Buffer Representation

- **Ring Buffers:** The key to this asynchronous pattern is the ring buffer (represented in the figure 3.3 as the arrows marked with a circle), which decouples the Host from the Coordinator, designed ad-hoc for the scheduler, this structure enables extremely fast communication with minimal overhead, using atomic variables instead of locks. As the graph illustrates, this mechanism also serve the communication between the Coordinator and the Workers, minimizing overhead and maximizing speed during task distribution. More on the code implementation in chapter ??.

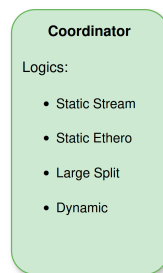


Figure 3.3: Coordinator

- **Coordinator:** The Coordinator acts as the brain of the scheduler, it decides how to dispatch tasks and implements different strategies to demonstrate various resource management techniques.

It features distinct scheduling logics, based on the type of workload that the host have to compute, the main logics that shape the design of the scheduler are the static one.

The first logic, Static stream, handles static homogeneous workloads (tasks of identical matrices), by using a Performance Map (a custom data structure containing heuristics and execution time estimates) the coordinator predicts how long a task will take on a specific accelerator and based on this, it performs a static dispatch of the tasks in a batch style manner (see section static partitioning ??).

The second logic, Static Hetero, work in a similar way but for static heterogeneous workloads (matrices of different sizes), the coordinator uses the same performance map to calculate the optimal distribution, ensuring the workload is balanced across all accelerators (see section static hetero partitioning ??).

To evaluate these strategies, we also implemented a dynamic tasks dispatch logic. This approach assigns tasks to the first available Worker (in idle state) and serves as a baseline to verify the efficiency of our static methods (see section dynamic ??).

Finally, the Coordinator supports the logic of Task Splitting, if a matrix exceeds the memory capacity of a single accelerator, the logic breaks it down into smaller sub-tasks to be distributed across multiple accelerators (see section large matrix split ??).

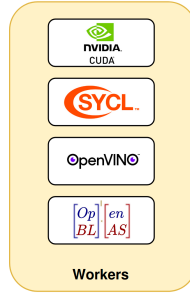


Figure 3.4: workers

- **Workers:** Workers are threads that utilize the abstraction layers defined in the preliminary phase 3.1. Their internal logic is straightforward: they retrieve available tasks and execute them on their assigned accelerator. All low-level details regarding the accelerator libraries are completely hidden from the Worker prospective (see Section ??).

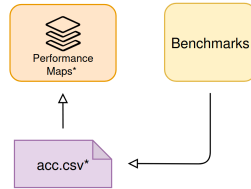


Figure 3.5: benchmark and performance maps rapresentation

- **Benchmarks and Performance Maps:** Internally, the scheduler utilizes a data structure called the Performance Map to determine the execution time of every single task for each individual accelerator. To achieve this, the structure contains heuristics and data loaded from CSV files (specifically one file for each accelerator). These files are created by a benchmark routine developed directly within the scheduler; if the csv for one or more accellerators is missing so that there are no data for the performance maps, the routine activates automatically to generate the necessary information stored in the csv files, it utilizes the accelerator libraries, testing them by executing precise tasks in a specific manner to gather this data as accurately as possible. Furthermore, the Performance

Map is an extremely fast structure: it is equipped with caching and highly optimized internal data structures to minimize overhead on the Coordinator's logic, which is the only component that accesses this structure (see Section ??).

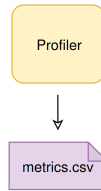


Figure 3.6: profiler

- **Profiler:** To conduct an extensive and precise study of how the scheduler behaves across all workloads, a component called the Profiler was developed directly within the scheduler. It is responsible for profiling all operational strategies and recording all execution times for both the scheduler and the workers. It returns all these metrics in an organized and readable manner in the form of CSV file. These metrics include the power consumption of the entire system and of each individual accelerator, as well as the specific overhead for every single phase of the scheduler's logic and a lot of more information that will be described later (see Section ??).

Chapter 4

Implementation

In this chapter, after having seen a general explanation of how the scheduler works in the previous sections, we are going to look at and explain all its components in detail.

In the first section, we will explain the structure of the project files and the reason why they are organized this way, and we will also give a view of the entire compilation process of the scheduler.

Subsequently, we will explain each of the abstraction layers used to target the accelerators, and then to conclude, we will see in detail all the components that make the scheduler logic work.

As mentioned in Chapter 3, the code developed during the preliminary phase is not analyzed in this section. This avoids repetition, as the evolved version of this code, which is integrated into the scheduler, will be explained in detail later. However, the original code is available in Appendix ??.

4.1 Project Structure and Compilation

The figure below provides a view of the project's file and folder structure:



Figure 4.1: Project Dir Tree

The codebase of this project was designed with a specific goal: modularity. Since the project works with very different hardware architectures, writing all the code in a single environment is not feasible. Each accelerator requires its own specific compiler and libraries: NVIDIA GPUs need the `nvcc` compiler, Intel GPUs need the `icpx` compiler for SYCL, and the NPU relies on OpenVINO libraries.

To solve this compatibility issue, the architecture was decomposed into independent dynamic libraries. Each hardware backend (CUDA, SYCL, OpenVINO) is compiled into a separate shared object file (`.so`), and each of these libraries expose a standard C ABI (Application Binary Interface). This design is mandatory because the C ABI acts as a universal language that allows the main scheduler to communicate with the accelerators without needing to understand their specific C++ implementations or complex driver dependencies, moreover the real problem was with the version of the compilers utilized with each specific library, so the standard approach is to unify all the interfaces with the help of the C language.

As shown in Figure 4.1, the project includes a dedicated folder for each library (`sycl/`, `openvino/`, and `cuda/`) containing their specific implementations, finally, the `scheduler/` directory containing all the internal dependencies used by the scheduler logic, which will be explained in section ??.

4.1.1 Compilation Process

The tool CMake was used in this project to manage this complex build process, the build is not performed all at once.

Below is the CMakeLists.txt file in the root directory:

```

1 cmake_minimum_required(VERSION 3.15)
2 project(amm_scheduler)
3
4 include(ExternalProject)
5
6 set(LIB_OUTPUT_DIR ${CMAKE_SOURCE_DIR}/bin)
7 file(MAKE_DIRECTORY ${LIB_OUTPUT_DIR})
8
9 # CUDA SYCL OPENVINO
10 option(USE_CUDA "CUDA ON" OFF)
11 option(USE_SYCL "SYCL ON" OFF)
12 option(USE_OPENVINO "OpenVINO ON" OFF)
13 option(USE_OPENBLAS "OpenBLAS ON" OFF)
14 option(USE_OPENCV "OpenCV ON" OFF)
15
16 set(SCHEDULER_DEPS "") # depends variable
17
18 set(OPT_FLAGS "-O2 -DNDEBUG")
19
20 # CUDA

```

```
21 if(USE_CUDA)
22     message(STATUS "CUDA Backend ON")
23     ExternalProject_Add(build_cuda
24         SOURCE_DIR ${CMAKE_SOURCE_DIR}/cuda
25         CMAKE_ARGS
26             -DCMAKE_INSTALL_PREFIX=${LIB_OUTPUT_DIR}
27             -DCMAKE_BUILD_TYPE=Release
28             -DCMAKE_EXPORT_COMPILE_COMMANDS=ON
29             -DCMAKE_CXX_FLAGS_RELEASE=${OPT_FLAGS}
30             -DCMAKE_CUDA_FLAGS_RELEASE=${OPT_FLAGS}
31             #BUILD_ALWAYS 1
32     )
33
34     list(APPEND SCHEDULER_DEPS build_cuda)
35 else()
36     message(STATUS "CUDA Backend OFF")
37 endif()
38
39 # SYCL
40 if(USE_SYCL)
41     message(STATUS "SYCL Backend: ON")
42     ExternalProject_Add(build_sycl
43         SOURCE_DIR ${CMAKE_SOURCE_DIR}/sycl
44         CMAKE_ARGS
45             -DCMAKE_INSTALL_PREFIX=${LIB_OUTPUT_DIR}
46             -DCMAKE_BUILD_TYPE=Release
47             -DCMAKE_EXPORT_COMPILE_COMMANDS=ON
48             -DCMAKE_CXX_COMPILER=icpx
49             -DCMAKE_CXX_FLAGS_RELEASE=${OPT_FLAGS}
50             #BUILD_ALWAYS 1
51     )
52     list(APPEND SCHEDULER_DEPS build_sycl)
53 else()
54     message(STATUS "SYCL Backend: OFF")
55 endif()
56
57 # OPENVINO
58 if(USE_OPENVINO)
59     message(STATUS "OPENVINO Backend: ON")
60     ExternalProject_Add(build_openvino
61         SOURCE_DIR ${CMAKE_SOURCE_DIR}/openvino
62         CMAKE_ARGS
63             -DCMAKE_INSTALL_PREFIX=${LIB_OUTPUT_DIR}
64             -DCMAKE_BUILD_TYPE=Release
65             -DCMAKE_EXPORT_COMPILE_COMMANDS=ON
66             -DCMAKE_CXX_FLAGS_RELEASE=${OPT_FLAGS}
67             #BUILD_ALWAYS 1
68     )
69     list(APPEND SCHEDULER_DEPS build_openvino)
70 else()
71     message(STATUS "OPENVINO Backend: OFF")
72 endif()
```

```

73
74 # scheduler
75 ExternalProject_Add(build_scheduler
76     SOURCE_DIR ${CMAKE_SOURCE_DIR}/scheduler
77     CMAKE_ARGS
78         -DCMAKE_INSTALL_PREFIX=${LIB_OUTPUT_DIR}
79         -DCMAKE_BUILD_TYPE=Release
80         -DLIBS_DIR=${LIB_OUTPUT_DIR}
81         -DCMAKE_CXX_FLAGS_RELEASE=${OPT_FLAGS}
82
83         -DCUDA_SRC_DIR=${CMAKE_SOURCE_DIR}/cuda
84         -DSYCL_SRC_DIR=${CMAKE_SOURCE_DIR}/sycl
85         -DOPEN_SRC_DIR=${CMAKE_SOURCE_DIR}/openvino
86
87         -DENABLE_CUDA=${USE_CUDA} # USE_CUDA
88         -DENABLE_SYCL=${USE_SYCL} # USE_SYCL
89         -DENABLE_OPENVINO=${USE_OPENVINO} # USE_OPENVINO
90         -DENABLE_OPENBLAS=${USE_OPENBLAS} # USE_OPENBLAS
91         -DENABLE_OPENCV=${USE_OPENCV} # USE_OPENCV
92
93         -DCMAKE_EXPORT_COMPILE_COMMANDS=ON # for clangd
94     DEPENDS ${SCHEDULER_DEPS}
95     #BUILD_ALWAYS 1
96 )
97
98 add_custom_target(UpdateLSP
99     ALL
100     COMMAND ${CMAKE_COMMAND} -E echo "update LSP..."
101     COMMAND python3 ${CMAKE_SOURCE_DIR}/merge_clangd.py
102     COMMENT "merge compile_commands.json for clangd"
103     DEPENDS build_scheduler
104 )

```

Listing 4.1: CMakeLists.txt root directory

As the code in the figure 4.1.1 show, the `ExternalProject_Add` feature of cmake is utilized to keep the compilation steps isolated. The main `CMakeLists.txt` file in the root directory acts as a manager. It does not compile the code directly, instead, it checks the configuration flags (such as `USE_CUDA`, `USE_SYCL` and `USE_OPENVINO`) and triggers a separate build process for each active module. The process is the following:

1. First, CMake enters the accelerators sub-directories, `cuda/`, `sycl/`, `openvino/`, and builds them as standalone projects.
2. Each sub-project uses its specific compiler to generate a dynamic library, e.g., `libcuda_lib.so`, `libsycl_lib.so`).
3. Finally, the `scheduler/` directory is compiled. Because of the standard C ABI, the scheduler does not need to know how the hardware works or which compiler

was used for the accelerators, it simply links against these shared libraries at runtime.

This separation ensures that the specific code for one hardware component does not interfere with the others, for example, the scheduler compiler does not need to handle CUDA Kernels, it only sees standard C functions.

Based on the flags passed to CMake during compilation, specific preprocessor definitions are passed to the C++ code. This allows the program to enable or disable dependencies on specific libraries directly inside the source code.

Consequently, the scheduler logic knows exactly which accelerators are available. This approach makes the system flexible, if a specific accelerator or library is missing on the current machine where the project have to be compiled, it can be disabled simply by using a flag.

Below the following snippet from the scheduler's build configuration shows how the flags trigger the injection of preprocessor definitions and the linking of specific libraries:

```

1 # CUDA
2 if(NOT DEFINED ENABLE_CUDA)
3     message("[SCHEDULER] CUDA off on scheduler")
4     set(ENABLE_CUDA OFF)
5 endif()
6
7 # SYCL
8 if(NOT DEFINED ENABLE_SYCL)
9     message("[SCHEDULER] SYCL off on scheduler")
10    set(ENABLE_SYCL OFF)
11 endif()
12
13 # OPENVINO
14 if(NOT DEFINED ENABLE_OPENVINO)
15    set(ENABLE_OPENVINO OFF)
16 endif()
17
18 #OPENBLAS
19 if(NOT DEFINED ENABLE_OPENBLAS)
20    message("[SCHEDULER] OpenBLAS off on scheduler")
21    set(ENABLE_OPENBLAS OFF)
22 endif()

```

Listing 4.2: Part of CMakeLists.txt scheduler directory

The command `target_compile_definitions` passes a macro (e.g., `ENABLE_CUDA`) to the compiler, which the C++ code effectively uses to enable or disable code blocks via `#ifdef` directives. Simultaneously, `target_link_libraries` ensures that the scheduler only links against the dynamic libraries that are actually present.

The compilation process for the accelerator backends follows a standard pattern. As an example, the configuration file for the OpenVINO library is presented below:

```

1 cmake_minimum_required(VERSION 3.20)
2 project(OpenVinoPart LANGUAGES CXX)
3
4 message("\n----- OPENVINO COMPILING \n")
5
6 find_package(OpenVINO REQUIRED)
7
8 add_library(ov_lib SHARED runner.cpp)
9
10 set_target_properties(ov_lib PROPERTIES PUBLIC_HEADER
    ov_wrapper.h)
11
12 # C++17 needed OpenVINO
13 target_compile_features(ov_lib PRIVATE cxx_std_17)
14
15 target_link_libraries(ov_lib PRIVATE openvino::runtime)
16
17 install(TARGETS ov_lib
18     LIBRARY DESTINATION lib
19     PUBLIC_HEADER DESTINATION include
20 )

```

Listing 4.3: CMakeLists.txt OpenVINO directory

First this script locates the necessary dependencies on the host system using `find_package`, then, it defines a shared library named `ov_lib` using the source file `runner.cpp`. The `SHARED` keyword is fundamental because it instructs the compiler to generate a dynamic library instead of a standard executable.

Finally, the script links the specific runtime (in this case, `openvino::runtime`) and specifies the installation rules. The `install` command ensures that the resulting binary and the public header (`ov_wrapper.h`) are placed in the correct output directories, making them accessible to the main scheduler.

The CUDA and SYCL modules utilize an identical structure, differing only in the specific compilers and libraries linked.

Finally, when the compilation finishes, all the resulting files are placed in a `bin/` directory. This folder contains the final executable named `scheduler`, the `csv` that the scheduler will create at runtime, and all the dynamic libraries it needs to run.

To speed up the development process, a script named `make_run.sh` was created. This script automates the entire workflow that consist of cleaning the `bin` and `build` dir, runs CMake with the correct options to enable all accelerators, compiles the project using all available CPU cores, and finally executes the program.

Additionally, the project includes a Python script named `merge_clangd.py`. Since

multiple independent builds are used, the LSP of the code editing tools often don't work with these many libraries, this script collects the compilation commands from all the different modules and merges them into a single file, ensuring that code completion and error highlighting work correctly across the entire project. This file will be executed after all the compilation steps are done.

4.2 Accelerators Abstraction Layers

Before diving into the architectural details in the following sections, it is important to clarify the supported data types. The scheduler is fully capable of managing both 32-bit and 16-bit matrix operations, furthermore thanks to the modular and flexible design of the codebase, the architecture is already provisioned to support 8-bit matrices as well. However, as 8-bit operations are not supported by all target accelerators and were not required to achieve the objectives of this thesis, their implementation has not been finalized across all backends.

4.2.1 Cuda

The CUDA abstraction layer is designed to manage the NVIDIA discrete GPU available in the system. As previously discussed, the interaction between the C++ scheduler and the NVIDIA compiler (`nvcc`) is mediated by a C interface.

The header file `cuda_wrapper.h` defines the Application Binary Interface exposed to the scheduler. It declares the necessary functions for initialization, memory cleanup, and the execution of matrix multiplication across different data types, 32-bit floating point, 16-bit floating point, and 8-bit integer.

```

1  #ifdef __cplusplus
2  extern "C" {
3  #endif
4
5      void cuda_init();
6      void cuda_free();
7      void cuda_gemm_32bit(void* A, void* B, void* C, int M, int
      N, int K);
8      void cuda_gemm_16bit(void* A, void* B, void* C, int M, int
      N, int K);
9      void cuda_gemm_8bit(void* A, void* B, void* C, int M, int
      N, int K);
10
11 #ifdef __cplusplus
12 }
13 #endif

```

Listing 4.4: `cuda_wrapper.h` interface

The implementation of these functions is in the `runner.cu` file, but before analyzing the initialization and execution logic, we will examine the global data structures and helper macros defined in the file, these elements manage the persistent state of the device and ensure error reporting.

```

1  #define CHECK_CUDA(func) {                                \
2      cudaError_t e = (func);                                \
3      if(e != cudaSuccess)                                    \
4          printf ("%s %d CUDA ERROR: %s\n", __FILE__, __LINE__, \
5                  cudaGetErrorString(e)); \
6  }
7  #define CHECK_CUBLAS(func) {                                \
8      cublasStatus_t e = (func);                                \
9      if(e != CUBLAS_STATUS_SUCCESS)                            \
10         printf ("%s %d CUBLAS ERROR: ", __FILE__, __LINE__); \
11 }
12 #define N_STREAM 2
13
14 cublasHandle_t handle;
15 cudaStream_t streams[N_STREAM];
16 void *d_A[N_STREAM];
17 void *d_B[N_STREAM];
18 void *d_C[N_STREAM];
19 int i = 0; /* stream id*/

```

Listing 4.5: cuda_runner.cu data and helper

To ensure that the execution goes without errors, the code utilizes two macros: `CHECK_CUDA` and `CHECK_CUBLAS`. These are used to wrap every API call to the CUDA runtime and the cuBLAS library, respectively, if an operation fails the macro immediately prints the specific error code along with the file name and line number where the failure occurred.

Regarding the state management, the system relies on CUDA Streams. A macro `N_STREAM` defines the number of parallel execution queues. Streams, enable the GPU to allows the overlap of memory transfers with computation, or the concurrent execution of multiple kernels. To support this concurrency, the device resources are replicated for each stream. The memory pointers (`d_A`, `d_B`, `d_C`) are defined as arrays, meaning that each stream possesses its own independent pre-allocated memory buffers. Finally, a global integer `i` acts as counter, used to cycle through the available streams when dispatching new tasks.

The design focuses on minimizing the overhead during the runtime by performing expensive operations only once during the startup. The `cuda_init` function is responsible for preparing the GPU environment before any task is processed. Its primary goal is to eliminate the latency associated with running the dynamic memory allocation istructions. Allocating memory on the GPU using `cudaMalloc` is a slow operation compare to the others, infact if the system were to allocate and free memory for every single matrix multiplication task, the overhead would significantly impact the performance. To solve this, the initialization function adopts a simple pre-allocation strategy. Below the implementation of the `cuda_init` function:

```

1 void cuda_init() {
2     CHECK_CUBLAS(cublasCreate(&handle));
3
4     /* reset stream counter */
5     i = 0;
6
7     size_t MAX = (size_t)4096 * 4096 * sizeof(float);
8
9     for (int j = 0; j < N_STREAM; ++j) {
10        CHECK_CUDA(cudaStreamCreate(&streams[j]));
11        CHECK_CUDA(cudaMalloc(&d_A[j], MAX));
12        CHECK_CUDA(cudaMalloc(&d_B[j], MAX));
13        CHECK_CUDA(cudaMalloc(&d_C[j], MAX));
14    }
15 }

```

Listing 4.6: cuda_runner.cu init

At startup, the function first initializes the cuBLAS library handle, which is required to invoke the linear algebra routines. Then, it iterates through `N_STREAM` number of streams (simply 2 stream is enough). For each stream, it creates a CUDA stream object to manage execution queues and allocates three distinct memory buffers on the device (`d_A`, `d_B`, `d_C`). The size of these buffers is calculated to hold the largest matrix dimension supported by the system (MAX size), ensuring that any incoming task fits within the pre-allocated space without requiring reallocation.

It is important to note that, although the accelerator hardware supports dimensions exceeding this limit, a maximum square matrix size of 4096×4096 was established (for each accelerator). This design choice was made to avoid introducing unnecessary architectural complexity, since it wasn't in this thesis objectives to maximize the memory saturation, and this limit provides a sufficient workload to stress-test the scheduler without complicating the memory management layer.

Complementary to the initialization, the `cuda_free` function ensures the proper release of all GPU resources when the scheduler shuts down:

```

1 void cuda_free() {
2     for (int j = 0; j < N_STREAM; ++j) {
3         CHECK_CUDA(cudaFree(d_A[j]));
4         CHECK_CUDA(cudaFree(d_B[j]));
5         CHECK_CUDA(cudaFree(d_C[j]));
6         CHECK_CUDA(cudaStreamDestroy(streams[j]));
7     }
8     CHECK_CUBLAS(cublasDestroy(handle));
9 }

```

Listing 4.7: cuda_runner.cu free

The function iterates through the created streams, freeing the device memory buffers using `cudaFree` and destroying the stream objects with `cudaStreamDestroy`. Finally, it destroys the global cuBLAS handle to release the library resources.

To maintain compatibility between the C ABI and CUDA C++ features, the execution logic is split into two levels, since the public C interface cannot recognize CUDA specific types it relies on generic `void*` pointers.

To handle this the system implements a layer of simple bridge functions, these functions accept the generic pointers and cast them to the correct concrete type (e.g., `float*` or `__half*`), and immediately forward the call to the internal function.

```

1 void cuda_gemm_32bit(void* A, void* B, void* C, int M, int N,
   int K) {
2     cuda_gemm_32bit_p((float*)A, (float*)B, (float*)C, M, N, K
   );
3 }
4
5 void cuda_gemm_16bit(void* A, void* B, void* C, int M, int N,
   int K) {
6     cuda_gemm_16bit_p((__half*)A, (__half*)B, (__half*)C, M, N
   , K);
7 }

```

Listing 4.8: `cuda_runner.cu` type bridging functions

The actual computational logic is implemented in the private functions, such as `cuda_gemm_32bit_p`. This functions orchestrates the entire lifecycle of a single GPU task.

```

1 void cuda_gemm_32bit_p(float* A, float* B, float* C, int M,
   int N, int K) {
2
3     cublasSetStream(handle, streams[i]);
4
5     CHECK_CUDA(cudaMemcpyAsync(d_A[i], A, M * K * sizeof(
   float), cudaMemcpyHostToDevice, streams[i]));
6     CHECK_CUDA(cudaMemcpyAsync(d_B[i], B, K * N * sizeof(
   float), cudaMemcpyHostToDevice, streams[i]));
7
8     float a = 1.0f, b = 0.0f;
9
10    cublasGemmEx(handle, CUBLAS_OP_N, CUBLAS_OP_N,
11        N, M, K,
12        &a,
13        d_B[i], CUDA_R_32F, N,
14        d_A[i], CUDA_R_32F, K,
15        &b,
16        d_C[i], CUDA_R_32F, N,

```

```
17         CUBLAS_COMPUTE_32F ,
18         CUBLAS_GEMM_DEFAULT_TENSOR_OP);
19
20     CHECK_CUDA(cudaMemcpyAsync(C, d_C[i], M * N * sizeof(
21     float), cudaMemcpyDeviceToHost, streams[i]));
21     CHECK_CUDA(cudaStreamSynchronize(streams[i])); /*
22     blocking for latency metrics */
22
23     i = (i + 1) % N_STREAM; /* change stream for the next
24     call */
24 }
```

Listing 4.9: cuda_runner.cu 32-bit execution logic

The function begins by associating the global cuBLAS handle with the current stream (`streams[i]`), this ensures that all subsequent library calls are queued into this specific stream.

The execution begins by asynchronously copying the inputs A and B to the GPU buffers using the `cudaMemcpyAsync`, next, the matrix multiplication is triggered via `cublasGemmEx`, configured to utilize Tensor Cores. Once computed, the result is copied back to the host. The process concludes with `cudaStreamSynchronize`, which blocks the thread to ensure data validity and accurate timing, followed by an update of the stream index for the next stream.

Chapter 5

Results

Appendix A

Questa è un'appendice

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Bibliography

- [1] Intel Corporation. *Intel Arc Graphics Xe-LPG Architecture*, 2023.
<https://www.intel.com/content/www/us/en/docs/oneapi/optimization-guide-gpu/2023-2/intel-xe-gpu-architecture.html>.
- [2] Intel Corporation. *Intel Core Ultra Processors (Series 1)*, 2024.
<https://www.intel.com/content/www/us/en/products/sku/236849/intel-core-ultra-9-processor-185h-24m-cache-up-to-5-10-ghz/specifications.html>.
- [3] Intel Corporation. Intel oneapi: A unified programming model, 2024.
<https://www.intel.com/content/www/us/en/developer/tools/oneapi/overview.html>.
- [4] Intel Corporation. *Intel's Neural Processing Unit*, 2024.
- [5] Intel Corporation. *Openvino Documentation*, 2025.
<https://docs.openvino.ai/2025/index.htm>.
- [6] Brian Kernighan and Dennis Ritchie. *The C Programming Language*. Prentice Hall, 1988.
- [7] Kitware, Inc. *CMake Documentation*, 2025.
<https://cmake.org/cmake/help/latest/>.
- [8] Klaus Hinum. *NVIDIA GeForce RTX 4060 Laptop GPU Spec*, 2022.
<https://www.notebookcheck.net/NVIDIA-GeForce-RTX-4060-Laptop-GPU-Benchmarks-and-Specs.675692.0.html>.
- [9] NVIDIA Corporation. *Cublas*, 2025. <https://docs.nvidia.com/cuda/cublas/index.html>.
- [10] NVIDIA Corporation. *CUDA C++ Programming Guide*, 2025.
<https://docs.nvidia.com/cuda/cuda-c-programming-guide/>.
- [11] NVIDIA Corporation. *NVIDIA Management Library (NVML)*, 2026.
<https://developer.nvidia.com/management-library-nvml>.
- [12] OpenBLAS Team. *Openblas github*, 2024.
<https://github.com/OpenMathLib/OpenBLAS>.

-
- [13] Steve Rennich. *CUDA C++ Streams and Concurrency*, 2024.
<https://developer.download.nvidia.com/CUDA/training/StreamsAndConcurrencyWebinar.pdf>.
- [14] Bjarne Stroustrup. *The C++ Programming Language*. Addison-Wesley, 2013.
- [15] The Khronos Group. *Sycl overview and specification*, 2024.
<https://www.khronos.org/sycl/>.
- [16] Zhang Xianyi. Openblas, 2025. <http://www.openmathlib.org/OpenBLAS/>.